

Kernel Particle Filter for Visual Tracking

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Abstract—A new particle filter—the Kernel Particle Filter (KPF)—is proposed for visual tracking in image sequences. The KPF invokes kernels to form a continuous estimate of the posterior density function. Particles are allocated based on the gradient information estimated from the kernel density estimate of the posterior. Results from simulations and experiments with real video data show the improved performance of the proposed algorithm when compared with that of the standard particle filter. The superior performance is evident in scenarios of small system noise or weak dynamic models where the standard particle filter usually fails.

Index Terms—Bootstrap filter, kernel density estimation, mean shift, particle filter, target tracking.

I. INTRODUCTION

PARTICLE FILTERS (PFs) have gained special attention in diverse areas such as signal processing, wireless communication, robotics, and physics, due mainly to their ability to handle multimodal probability density functions (PDFs) [1]. In computer vision, the superior performance of PFs over Kalman filters in tracking objects in heavy clutter has been reported in [2] and by many other research groups thereafter. Despite its success in various applications, it was observed that a PF does not perform well when the dynamic system has a very small system noise or if the observation noise has very small variance. In these cases, the particle set quickly collapses to one single point in the state space, and the filter performance is severely affected. Improved particle filters have been proposed to address these issues. The Auxiliary Particle Filter (APF) [3], Likelihood Particle Filter (LPF) [4], and Regularized Particle Filter (RPF) [5] are three such examples. A survey of the most commonly used PFs can be found in [4].

Most of the improved particle filters cannot, however, be directly applied to visual tracking, where the likelihood PDF is not directly available [2]. This precludes the application of methods like LPF. On the other hand, it is often advantageous to have a small and fixed computation load at each time step. Methods that use a large (and/or variable) number of particles, such as RPF and APF [5], are less desirable in this case. Furthermore, in visual tracking, the tracked object (often a human subject or a body part) may sporadically perform motions that deviate from the presumed motion model. Even with a reasonably large process noise, the standard PF often fails to produce a particle

set that captures the “irregular” motion, leading to gradually drifting estimates and the ultimate loss of the target.

In this letter, we propose a new particle filter: the Kernel Particle Filter (KPF). KPF is similar to RPF in the sense that a kernel density estimate (KDE) [6] is used to approximate the posterior PDF. However, unlike RPF, which uses samples from the KDE to replace the original particles, KPF estimates the gradient of the kernel density and moves particles toward the modes of the posterior, leading to a more effective allocation of particles. The gradient estimation and particle allocation is implemented by the mean shift algorithm [7], [8]. We compare KPF with the standard PF using simulated and real data and show that KPF performs robust visual tracking with improved sampling efficiency.

II. KERNEL-BASED POSTERIOR ESTIMATION

A. KDE

Denote the target state and the observation at (discrete) time t as $\mathbf{x}_t$ and $\mathbf{y}_t$, respectively, where $\mathbf{x}_t \in \mathbb{R}^{n_x}$, $\mathbf{y}_t \in \mathbb{R}^{n_y}$. Let $\mathbf{Y}_t = \{\mathbf{y}_0, \dots, \mathbf{y}_t\}$ be the history of observations up to time t . In recursive Bayesian estimation, the posterior PDF is estimated by propagating the PDF over time:

$$p(\mathbf{x}_t | \mathbf{Y}_t) \propto p(\mathbf{y}_t | \mathbf{x}_t) p(\mathbf{x}_t | \mathbf{Y}_{t-1}). \quad (1)$$

KDE [6] is used in this work to form a continuous estimate of the posterior in order to facilitate gradient estimation. Given a particle set at time t : $\mathbf{S}_t = \{\mathbf{S}_t^{(n)}\}_{n=1}^N$ and associated weights $\{w_t^{(n)}\}_{n=1}^N$, the kernel density estimation of the posterior with kernel K can be formulated as

$$\hat{p}(\mathbf{x}_t | \mathbf{Y}_t) = \sum_{n=1}^N K_\lambda \left(\mathbf{x}_t - \mathbf{s}_t^{(n)} \right) w_t^{(n)} \quad (2)$$

where K_λ is the kernel scaled by the kernel width λ :

$$K_\lambda \left(\mathbf{x}_t - \mathbf{s}_t^{(n)} \right) = \frac{1}{\lambda^{n_x}} K \left(\frac{\mathbf{x}_t - \mathbf{s}_t^{(n)}}{\lambda} \right). \quad (3)$$

In essence, the kernel is used for interpolation, with each particle contributing to the estimate in accordance with its distance from $\mathbf{x}_t$. The kernel and width are chosen so as to minimize the Mean Integrated Square Error (MISE) between the posterior PDF and the corresponding kernel estimate. When a Gaussian kernel is used and the posterior is also Gaussian with a unit covariance matrix, the optimal kernel width is given by [6]

$$\lambda_{\text{opt}} = \left(\frac{4}{(n_x + 2)N} \right)^{\frac{1}{n_x + 4}}. \quad (4)$$

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Although (4) is optimal in the L^2 sense only for equally weighted particles and Gaussian density, it still can be used in the general case to obtain a suboptimal filter [5]. In practice, when densities are often multimodal, we let $\lambda = \lambda_0 = (1/2)\lambda_{opt}$ [5], [6].

B. Posterior Gradient Estimation

Given the posterior estimation (2), we now estimate its gradient and move particles along the gradient direction toward the modes of the posterior. This can be achieved using the mean shift procedure [7]. In this procedure, each particle is moved to its sample mean determined by

$$m(\mathbf{s}_t^{(n)}) = \frac{\sum_{l=1}^N H_\lambda(\mathbf{s}_t^{(n)} - \mathbf{s}_t^{(l)}) w_t^{(l)} \mathbf{s}_t^{(l)}}{\sum_{l=1}^N H_\lambda(\mathbf{s}_t^{(n)} - \mathbf{s}_t^{(l)}) w_t^{(l)}} \quad (5)$$

where H is an arbitrary kernel. It is shown that the mean shift vector $m(\mathbf{x}) - \mathbf{x}$ using kernel H would be in the gradient direction of (2) if the kernel profiles¹ h and k satisfy $h(r) = -ck'(r)$ for all $r \in [0, \infty)$ and some $c > 0$ [7]. Note that the derivative of a normal profile remains normal. It is also shown in [7] that mean shift is a steepest ascent procedure that will find *all* the local maxima of the KDE of the posterior.

Given $\mathbf{S}_t$ and the associated weights, the empirical covariance matrix $\hat{\mathbf{C}}_t$ of the particle set is computed first, and a “whitening” step is performed on each particle in which $\mathbf{s}_t^{(n)}$ is changed to $\mathbf{A}_t^{-1} \mathbf{s}_t^{(n)}$, where $\mathbf{A}_t \mathbf{A}_t^T = \hat{\mathbf{C}}_t$ to achieve a unit covariance matrix. A symmetric K_λ can then be used for mean shift, and the results are multiplied by $\mathbf{A}_t$.

III. KPF

A. Particle Reweighting

The mean shift can be applied repeatedly to a particle set. A problem arises when particles change their positions: The new particles do not follow the posterior distribution anymore. This is compensated in KPF by reweighting the particles. Denote the particle set after the i th mean shift procedure at time t as $\{\mathbf{s}_{t,i}^{(n)}\}$. After each mean shift procedure, the weight is recomputed as the posterior density evaluated at the new particle positions augmented with a particle density balancing factor

$$w_{t,i}^{(n)} = \frac{p(\mathbf{s}_{t,i}^{(n)} | \mathbf{Y}_t)}{q_{t,i}(\mathbf{s}_{t,i}^{(n)})} \quad (6)$$

where the denominator is the new proposal density that captures the inevitable nonuniformity of the new particle set

$$q_{t,i}(\mathbf{x}_t) = \sum_{l=1}^N K_\lambda(\mathbf{x}_t - \mathbf{s}_{t,i}^{(l)}) \quad (7)$$

and the posterior density is given by (1)

$$p(\mathbf{x}_t | \mathbf{Y}_t) \propto p(\mathbf{y}_t | \mathbf{x}_t) \cdot \sum_{l=1}^N p(\mathbf{x}_t | \mathbf{s}_{t-1}^{(l)}) w_{t-1}^{(l)}. \quad (8)$$

¹The *profile* of a kernel K is defined as a function $k: [0, \infty) \rightarrow \mathbb{R}$ such that $K(\mathbf{x}) = k(\|\mathbf{x}\|^2)$.

TABLE I
KERNEL PARTICLE FILTER ALGORITHM

Given $\{\mathbf{s}_{t-1}^{(\cdot)}, w_{t-1}^{(\cdot)}\}$:
$\mathbf{s}_{t,0}^{(\cdot)} = \text{Propagate}(\{\mathbf{s}_{t-1}^{(\cdot)}, w_{t-1}^{(\cdot)}\})$;
Calculate the empirical covariance matrix $\hat{\mathbf{C}}_t$ of $\{\mathbf{s}_{t,0}^{(\cdot)}\}$;
Compute $\mathbf{A}_t$ such that $\mathbf{A}_t \mathbf{A}_t^T = \hat{\mathbf{C}}_t$;
Draw $\mathbf{e}^{(\cdot)} \sim N(\mathbf{0}, \mathbf{I}_{n_x})$;
$\mathbf{s}_{t,1}^{(\cdot)} = \mathbf{s}_{t,0}^{(\cdot)} + \lambda_0 \cdot \mathbf{A}_t \cdot \mathbf{e}^{(\cdot)}$;
$w_{t,1}^{(\cdot)} = \text{Weight}(\mathbf{s}_{t,1}^{(\cdot)})$;
//Gradient ascent on the posterior:
For $i = 1 : I - 1$ // I : the number of iterations
$\mathbf{s}_{t,i+1}^{(\cdot)} = \text{MeanShift}(\{\mathbf{s}_{t,i}^{(\cdot)}, w_{t,i}^{(\cdot)}\})$; //Equation (5)
Draw $\mathbf{e}^{(\cdot)} \sim N(\mathbf{0}, \mathbf{I}_{n_x})$;
$\mathbf{s}_{t,i+1}^{(\cdot)} = \mathbf{s}_{t,i+1}^{(\cdot)} + \lambda_i \cdot \mathbf{A}_t \cdot \mathbf{e}^{(\cdot)}$;
$w_{t,i+1}^{(\cdot)} = \text{Reweight}(\mathbf{s}_{t,i+1}^{(\cdot)})$; //Equation (6)
end
$\hat{\mathbf{x}}_t = \sum_{n=1}^N \mathbf{s}_{t,I}^{(n)} w_{t,I}^{(n)}$;
$\mathbf{s}_t^{(\cdot)} = \mathbf{s}_{t,J}^{(\cdot)}$; $w_t^{(\cdot)} = w_{t,J}^{(\cdot)}$; ($0 \leq J \leq I$)

The second term on the right-hand side of (8) is a sample-based approximation of the prior density. When mean shift is performed on $\mathbf{S}_t$, the next generation of particles will concentrate more on the approximated density modes, but the weight will contain a factor that offsets this effect, allowing the next mean-shift iteration to follow the correct posterior gradient.

B. Implementation

A pseudo-code of the KPF algorithm at one time step is shown in Table I. It is found through pilot studies that in general, two to five iterations are sufficient to move particles close to the high probability areas in three to nine dimensional spaces. Note that to prevent local plateaus in the density from stopping the gradient ascent too early, a small perturbation is added to the particles at each iteration. While focusing particles on the modes of the density may impair the filter’s ability to carry multiple hypotheses, it is possible to alleviate the problem by propagating the particle set at the J th iteration, where $0 \leq J \leq I$, and I is the number of iterations, since at early stages of the iterations, the particle set is, in general, more scattered and carries more hypotheses. In our implementation, we let $J = I$. Rather than using a constant λ , it is more effective to use a wide kernel at first to smooth out certain weak modes in the PDF and gradually scale it down to move particles to the most dominant modes. In our experiments, we set $\lambda_i = \alpha^i \lambda_0$, where α is a number empirically chosen within the interval $[0.5, 1]$.

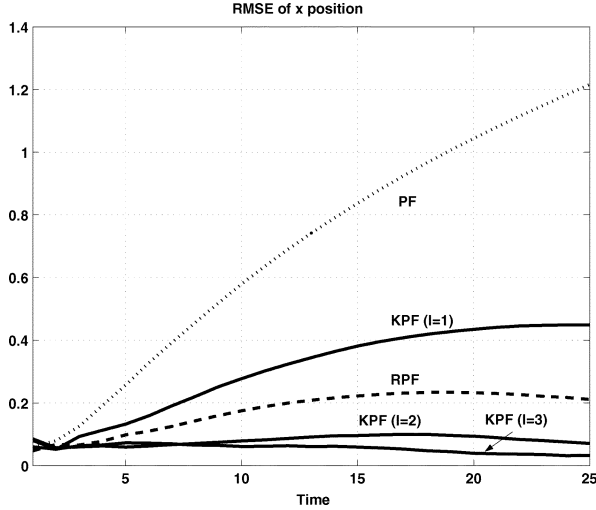


Fig. 1. Average RMSE of the estimated x position over 30 Monte Carlo runs on simulated data. The number of particles used is 600 particles for PF and RPF and $600/I$ for KPF with I iterations.

IV. EXPERIMENTAL RESULTS

Results on target tracking using both simulated data and real video are presented in this section.

A. Simulation Results

The KPF is found to provide better filtering performance on various dynamic systems compared with the standard PF, especially with the presence of system noise that is either too small or too large. Here, we present the simulation result on a standard constant velocity dynamic system with a small system noise. In this simulation, the target moves within a two-dimensional plane. Its trajectories are generated according to the dynamic model:

$$\mathbf{x}_t = \Phi \mathbf{x}_{t-1} + \Gamma \mathbf{w}_t \quad (9)$$

where $\mathbf{x}_t = (x, \dot{x}, y, \dot{y})^T$, $\mathbf{w}_t = (w_x, w_y)^T$, $\Phi = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$, and $\Gamma = \begin{pmatrix} 0.5 & 0 \\ 1 & 0 \\ 0 & 0.5 \\ 0 & 1 \end{pmatrix}$. The system noise $\mathbf{w}_t$ is a zero-mean Gaussian process with covariance matrix $\mathbf{Q} = q^2 \mathbf{I}_2$, where $\mathbf{I}_2$ denotes the 2×2 identity matrix. An observer measures the x and y location of the target

$$\mathbf{y}_t = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \mathbf{x}_t + \mathbf{v}_t \quad (10)$$

where $\mathbf{v}_t$ is the zero-mean Gaussian measurement noise with covariance $\mathbf{R} = r^2 \mathbf{I}_2$. The initial state vector is assumed to have a Gaussian distribution with known mean $\bar{\mathbf{x}}_0$ and covariance matrix $\mathbf{C}_0$.

A trajectory and associated measurements over 25 time steps are generated with the parameter values $q = 0.002$, $r = 0.05$, and the actual initial state $\mathbf{x}_0 = (5.3, 0.43, 4.5, -0.52)^T$. A prior distribution with parameters $\bar{\mathbf{x}}_0 = (5, 0.5, 5, -0.5)^T$ and $\mathbf{C}_0 = \text{diag}(1, 0.1, 1, 0.1)$ is used to generate the initial particle set for all particle filters. Parameter α is set to 1 in KPF. The average Root Mean Square Error (RMSE) of the estimated x positions in 30 runs using the standard PF, RPF, and KPF with

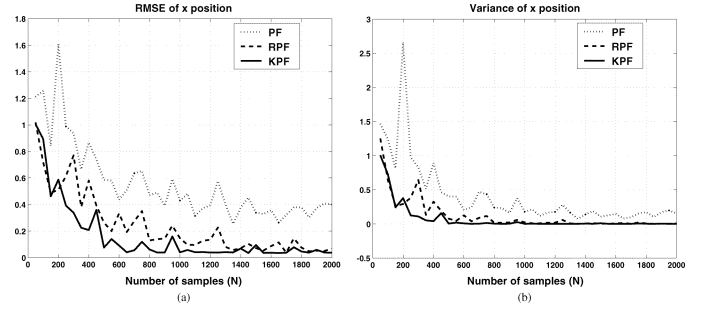


Fig. 2. Average (a) RMSE and (b) variance of the estimated x position with different number of particles N (30 runs on simulated data). The number of particles is $N/3$ for KPF with $I = 3$.

$N = 600$ particles is shown in Fig. 1 (results of estimated y positions and velocities are comparable to that of x). Because there are small “overlaps” between the high probability areas of the prior and the likelihood due to the small system noise, the standard PF with 600 particles quickly diverges. RPF is able to produce better results, while KPF with 200 samples and three iterations (600 weight computations) gives the best performance. The advantage of gradient ascent is evident in Fig. 1: The filter performance improves as the particles are brought closer to the high probability areas of the posterior.

Fig. 2 shows the average RMSE and variance of the estimated x position using the three filters with different numbers of particles. The results are averaged over 30 Monte Carlo runs. While the performance of all three filters improves when more particles are used, it is clear that KPF provides faster convergence.

B. Video Tracking Results

The KPF algorithm is applied to track moving human faces in various videos containing both indoor and outdoor scenes. Face motions in the videos include sudden acceleration, rotation, abrupt changes of direction, and jumping. We present the result on one of the test videos here.

The video sequence, captured by a zooming camera at 15 frames/sec with a resolution of 180×120 pixels, consists of 797 frames of a human face moving in a typical laboratory environment.² The face is modeled as an ellipse with a vertical major axis and a fixed aspect ratio of 1.4, so that the state is determined by three parameters $\mathbf{x}_t = (x, y, \sigma)^T$, where (x, y) is the position of the ellipse’s center and σ the length of the ellipse’s minor axis (in pixels). The weight of each sample is determined by the product of two factors based on color and gradient. The gradient factor is determined by the pixel values of the edge map on the perimeter of the model. For the color factor, a cosine similarity is computed between each hypothesis histogram and a model histogram obtained prior to tracking. The color space employed is the normalized $r - g$ space augmented with intensity I , with eight bins for each chromaticity channel and four bins for the intensity channel.

The experimental video considered contains various motions, such as horizontal acceleration, jumping, and out-of-plane rotation. A random walk motion model is used for both PF and KPF, i.e., $\mathbf{x}_t = \mathbf{x}_{t-1} + \mathbf{w}_t$. The noise $\mathbf{w}_t$ is assumed to be a Gaussian

²The test video can be accessed, along with other videos, from <http://ece.uic.edu/~cchang>.

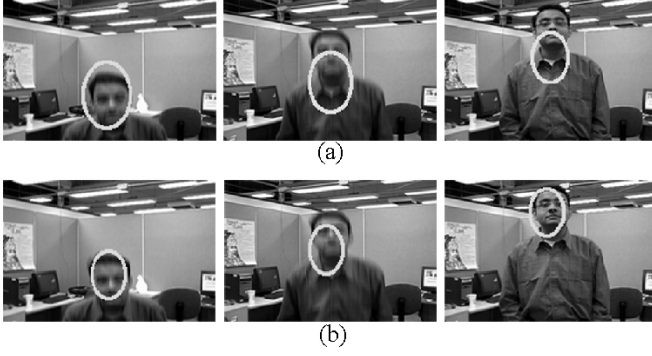


Fig. 3. Estimated ellipse in frames #24, #27, and #31 using (a) PF with 200 particles and (b) KPF with 30 particles and three iterations.

process with zero mean and covariance $\text{diag}(3^2, 3^2, 0.5^2)$. Parameter α is set to 0.8 in KPF. Trackers are initialized manually. A few frames of the tracking results using PF and KPF are shown in Fig. 3. The PF with 200 particles tends to lag behind when the weak dynamic model produces prior samples that fail to cover the likelihood modes.³ The PF tracker ultimately loses the head at the 373rd frame. On the other hand, KPF with 30 particles and three iterations, despite being occasionally distracted by the background clutter, is able to robustly track the face throughout the sequence. Fig. 4 shows the error of the estimated x and y position in the first 496 frames.

C. Computational Cost and Accuracy

KPF improves filtering performance at the cost of introducing extra computation. Equations (5), (7), and (8) all require $O(N^2)$ function evaluations. However, it is possible to greatly reduce the computation complexity to $O(N)$ by seeking suboptimal results. Note that the kernel evaluations in (5) and (7) need to be computed only once for every iteration if we use a normal kernel such that $H_\lambda = K_\lambda$. Also, the summation of N terms in the two equations may be reduced to fewer terms. In our experiments, it was possible to reduce the summation to ten terms from $N = 200$ without affecting the results significantly. A similar reduction can be made to (8). Computations can be further reduced when accurate initialization is available and the observation model is robust, in which case, the motion model may be relaxed by skipping the reweighting step.

In visual tracking, KPF can actually improve the filter efficiency by reducing the weight computation. Assume N_{pf} particles are used in PF and N_{kpf} in each iteration of KPF. If each sample weighting takes T_w , and each mean shift and reweighting takes T_{ms} , then $N_{\text{pf}} \cdot T_w$ are used by PF for each frame and $N_{\text{kpf}} I \cdot T_w + (I - 1) N_{\text{kpf}} \cdot T_{\text{ms}}$ by a KPF with I iterations. In our C++ implementation of the KPF algorithm, the algorithm works at 10 Hz for the video presented without any code optimization and $T_{\text{ms}} \approx T_w/2$. In this case, KPF will be less time consuming than PF, as long as $N_{\text{kpf}} < (1/4)N_{\text{pf}}$, assuming $I = 3$.

The computation comparison is only meaningful when both KPF and PF succeed in tracking. Our experiments showed that PF often fails in handling weak dynamic models. We have ob-

³It should be pointed out that a well-trained motion model may improve the PF's performance.

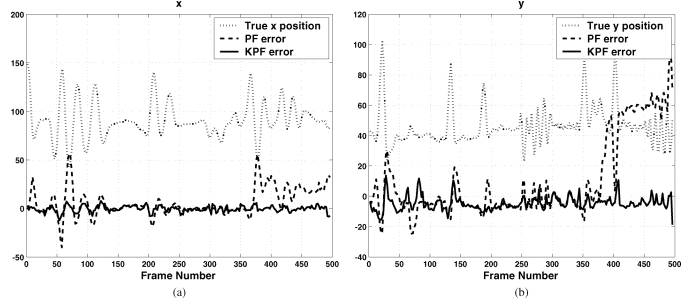


Fig. 4. Error of the estimated head centers in the first 496 frames, along with the true (hand-marked) head centers shown as dotted lines. (a) $\hat{x}_t - x_t$. (b) $\hat{y}_t - y_t$. The PF tends to lag behind as the head changes its moving direction and ultimately loses the head at the 373rd frame.

served cases where PF still diverges, even after N_{pf} is significantly increased. In situations where PF manages to succeed, it is illuminating to look at the achieved RMSE given the number of weight computations N_w ($N_w = N_{\text{pf}}$ for PF and $N_w = I \cdot N_{\text{kpf}}$ for KPF). Assume that the RMSE reduces exponentially with the increase of N_w , and let the RMSE be $Ae^{-\beta N_{\text{pf}}}$ for PF and $Ae^{-\gamma(I \cdot N_{\text{kpf}})}$ for KPF. The KPF will achieve smaller RMSE as long as $N_{\text{kpf}} > (\beta/\gamma \cdot I)N_{\text{pf}}$. Fitting the exponential functions to Fig. 2(a) using Least-Square estimates, we get $\beta = 0.001$ and $\gamma = 0.0022$. Therefore, for $I = 3$, the KPF will achieve better RMSE if $N_{\text{kpf}} > (1/6.6)N_{\text{pf}}$. This rough analysis indicates that it is possible to achieve better RMSE with less sample weighting using KPF, assuming that the PF will not fail in the first place.

V. CONCLUSIONS

We described a modified PF, the KPF, for tracking. The method is based on kernel-based density estimation of the posterior, with the mean shift algorithm serving as an efficient gradient estimation and mode-seeking procedure. The algorithm was applied to both simulated and real data and was found to provide improved tracking performance compared with the conventional PF. In the case of visual tracking in video, KPF further alleviates the computation burden by reducing the total number of weight computations.

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